choice in A at an input $c \in K \cap (\mathcal{M} \setminus \{0\})$ would have coordinates algebraic over K, t, c, hence in K by Lemma 2.13; their ratio cannot have leading term t. Even though finitary values may be empty, $b \in \lim g$ forces their support to accumulate at 0; by C-minimality that support contains a punctured neighborhood, whose defining radius lies in $\Gamma(K)$ since adjoining the residue parameter t does not enlarge definable values. It therefore contains such an input $c \in K$. Thus the claimed curve does not exist over that base. Impose VF-generation before this lemma (as already suggested in the human report), or allow explicitly specified extra VF representatives. Under VF-generation, the added nonzero $c \in \mathcal{M}$ makes the algebraic closure a model, and the displayed construction yields nonempty finite values converging to b itself. For the last rescaling step, first shrink to a disc whose radius is the valuation of a definable scalar; the original definable radius need not be realized by such a scalar.

Author resolution. The author approves VF-generation as a local hypothesis of this lemma, validating the algebraic-closure model argument. The continuous germ is extended by a constant finite nonempty set on a clopen complement, so no definable scalar realizing an arbitrary definable radius is assumed. Later geometric uses may temporarily enlarge the base; no global VF-generation assumption is imposed here.

This is an analogue of the highly useful o-minimal curve selection.

Remark 2.47. Although the Γ-sort is essentially an o-minimal group, curve selection fails due to the presence of the top element ∞. For example, the set $\Gamma \times (0, \infty) \subseteq \Gamma_\infty^2$ contains curves approaching the boundary point $(0, 0)$ or (∞, ∞) but no curves approaching the boundary point $(\infty, 0)$, because every definable one-parameter curve is piecewise rational-affine. Near a one-sided endpoint, one affine piece remains, and a positive affine coordinate cannot tend to 0 while another coordinate tends to ∞.

Resolved referee R166 [Typo; new] – Eventually affine curves in the value group.
Original source line 1094. Replace "all curves are lines" by "every definable one-parameter curve is piecewise rational-affine". Near a one-sided endpoint, one affine piece remains; a positive affine coordinate cannot tend to 0 while another coordinate tends to ∞. This is the precise reason the stated boundary point cannot be approached.

Author resolution. The author approves the precise piecewise rational-affine formulation. An eventual affine piece gives the stated obstruction to approaching $(\infty, 0)$; the example and the conclusion are unchanged.

Lemma 2.48. *Let $A \subseteq \mathrm{VF}^n$ be definable, closed and bounded, and let $f : A \longrightarrow \mathrm{VF}^m$ be definable and continuous. Then $f(A)$ is also closed and bounded.*

Proof. The conclusion is geometric, so we may temporarily enlarge the parameter base to a small VF-generated base over which all the data are definable. The standing nontriviality of the ground valuation is preserved. Over this base, and after naming any additional finite VF tuple, algebraic closure is a model of $\mathrm{ACVF}(0, 0)$. Elementarity and compactness therefore give finite nonempty definable choices in the nonempty families below. No choice over the original parameter base is asserted.

First let $b \in \overline{f(A)} \setminus f(A)$, where this closure is affine, and name b. For every $c \in \mathcal{M} \setminus \{0\}$ the set

$$\{x \in A : f(x) \in \mathfrak{o}(b, \mathrm{val}(c))\}$$

is nonempty. Applying the model argument over the enlarged base together with (b, c), and then compactness, gives a definable finitary function $F : \mathcal{M} \setminus \{0\} \longrightarrow \mathcal{P}(A)$ such that every $F(c)$ is nonempty and $f(F(c)) \subseteq \mathfrak{o}(b, \mathrm{val}(c))$. Lemma 2.45, in its finite-dimensional form, gives a finite minimal limit set $L \subseteq A$. This set is nonempty, since F has nonempty values arbitrarily close to 0. For every $a \in L$, minimality implies that values of F meet every neighborhood of a for inputs

arbitrarily close to 0: otherwise a could be removed from L. Continuity at a and the displayed containment force $f(a) = b$, contradicting $b \notin f(A)$. Thus $f(A)$ is closed.

If $f(A)$ were unbounded, one coordinate f_i would be unbounded. For every $c \in \mathcal{M} \setminus \{0\}$ the definable set

$$\{x \in A : \mathrm{val}(f_i(x)) < -\mathrm{val}(c)\}$$

would be nonempty. The same model argument and compactness give finite nonempty choices $F(c)$ inside these sets. Again Lemma 2.45 supplies a nonempty finite minimal limit set $L \subseteq A$. Choose $a \in L$. Continuity makes f_i bounded on a neighborhood of a in A, while minimality makes $F(c)$ meet that neighborhood for arbitrarily small nonzero c. The inequality $\mathrm{val}(f_i(x)) < -\mathrm{val}(c)$ for all $x \in F(c)$ contradicts this local bound. Hence $f(A)$ is bounded as well. □

Resolved referee R041 [Gap] – Selecting and lifting the limiting curve.
Original source line 1104. If the image is unbounded but closed in affine space, its affine frontier need not contain a point outside it. Take the frontier in the coordinatewise projective line and allow an infinite coordinate. Also the existence of algebraic points in every fiber needs justification under the chosen parameter assumptions; for the curve pullback it is enough to include the curve parameter c in the algebraic closure and apply the model argument there. Preserve a nonempty finite-valued parametrization throughout and apply the limit lemma in its multidimensional form.

Author resolution. The author approves treating affine missing boundary points and unbounded output separately. Finite nonempty choices are justified by elementarity and compactness over a temporary VF-generated base. The finite-dimensional limit theorem and continuity then give both closedness and boundedness, without a global VF-generation assumption or an unjustified original-base choice.

Corollary 2.49. *If the function f in Lemma 2.48 is injective then it is a homeomorphism from A onto $f(A)$.*

Lemma 2.50 (The closed graph criterion for continuity)**.** *Let $A \subseteq \mathrm{VF}^n$ and $B \subseteq \mathrm{VF}^m$ be definable, with B closed in affine space. Write B^{p} for the closure of B in $(\mathbb{P}^1(\mathrm{VF}))^m$, with the valuation topology. A definable function $f : A \longrightarrow B$ is continuous if and only if its graph is closed in $A \times B^{\mathrm{p}}$. If B is bounded, then $B^{\mathrm{p}} = B$, so this is the usual relative closed-graph condition.*

Proof. The graph of a continuous map into the Hausdorff space B^{p} is closed, proving the forward implication.

Conversely, suppose the graph is closed but f is not continuous at $a \in A$. Choose a clopen projective polydisc U around the finite point $f(a)$ such that

$$D = \{x \in A : f(x) \notin U\}$$

has a in its closure. As in the preceding proof, temporarily enlarge the parameter base to a small VF-generated base containing the parameters of these data. This changes neither continuity nor graph closure. By Lemma 2.46, there is a definable continuous finitary function $C : \mathcal{M} \setminus \{0\} \longrightarrow \mathcal{P}(D)$ with nonempty values and limit $\{a\}$.

Apply the projective version of Lemma 2.45 to the finite-valued function $c \longmapsto f(C(c))$. It has a finite minimal limit set L, which is nonempty because all the values are nonempty. Every point of L belongs to the closed set $B^{\mathrm{p}} \setminus U$: minimality makes the image meet each neighborhood of such a point for inputs arbitrarily close to 0. Choose $b' \in L$. This same property, together with $C(c) \longrightarrow \{a\}$, shows that (a, b') belongs to the closure of the graph of f in $A \times B^{\mathrm{p}}$. Closedness of the graph gives $b' = f(a)$, contrary to $b' \notin U$. This proves continuity without a global VF-generation hypothesis. □

Resolved referee R042 [Gap; revised] – The compactified closed-graph criterion.
Original source line 1117. The parenthesis appears intended to impose the correct compactified-target hypothesis; make it explicit by taking B to be a closed definable subset of $(\mathbb{P}^1(\mathrm{VF}))^m$, and take graph closure in $A \times B$. The example $f(0) = 0$, $f(x) = 1/x$ for $x \neq 0$ has closed graph in affine VF^2 but is discontinuous at 0: this is a counterexample only to the unbounded affine-target version, not to the intended compactified statement. For the reverse implication, apply curve selection to the set of input points x with $f(x) \notin \mathfrak{b}$, approaching a; apply the finite-limit lemma to the resulting finite-valued image in B. A limit point b' lies outside the clopen neighborhood $\mathfrak{b}$, and (a, b') lies in the graph closure. Also, "projective line with the valuation topology" is more precise than a literal topological one-point compactification; local compactness is not assumed here.

Author resolution. The author approves making the intended projective ambient space explicit: the affine target is B, and graph closure is in $A \times B^{\mathrm{P}}$, with the valuation topology. The reverse proof selects bad inputs first and then takes a finite projective limit of their images. Temporary base enlargement supplies curve selection without changing the geometric conclusion or imposing global VF-generation.

Lemma 2.51. *Let $f : A \longrightarrow B$ be a definable bijection between two open subsets of* VF*. Suppose that B is bounded. Then f is weakly concentric, that is, for every $a \in A$ there is a $b \in \mathrm{VF}$ such that, for every sufficiently large $\gamma \in \Gamma$, $f(\mathfrak{o}(a, \gamma) \smallsetminus a) = \mathfrak{o}(b, \delta) \smallsetminus b$ for some $\delta \in \Gamma$.*

Note that, compared to Lemma 2.32, the assumption of this lemma is stronger and its conclusion weaker, but the point is that we do not need a partition to achieve it.

Proof. By Lemma 2.45, for any $a \in A$, $\lim_{A \to a} f$ exists in the closure of B, which we denote by b_a. By C-minimality and o-minimality in the Γ-sort, $\mathfrak{c}(b_a, \delta) \smallsetminus \mathfrak{o}(b_a, \delta) \subseteq B$ for all sufficiently large $\delta \in \Gamma$. Also, for all sufficiently large $\gamma \in \Gamma$ there is a $\delta \in \Gamma$ such that

$$f(\mathfrak{c}(a, \gamma) \smallsetminus \mathfrak{o}(a, \gamma)) = \mathfrak{c}(b_a, \delta) \smallsetminus \mathfrak{o}(b_a, \delta)$$

The lemma follows. □

Referee R043 [Gap; revised] – The annulus equality needs a proof.
Original source line 1132. Specify that A and B each have one VF-coordinate. The finite limit is a singleton $\{b\}$ by [17, Lemma 9.4]; nonemptiness follows because the values are nonempty on a punctured neighborhood. The equality of whole thin annuli needs more explanation. Here is a repair. The valuation calculation in the proof of [17, Lemma 8.3], applied to $x \mapsto \mathrm{val}(f(x) - b)$, gives

$$\mathrm{val}(f(x) - b) = q\,\mathrm{val}(x - a) + c \qquad (q \in \mathbb{Q}_{>0})$$

on a sufficiently small punctured neighborhood: each unary polynomial has eventually monomial valuation, and the limit forces positive slope. Remove the at most one exceptional input with $f(x) = b$. By C-minimality, B contains a punctured neighborhood of b. The inverse has a unique projective limit there, necessarily a, since points $x \longrightarrow a$ give $f(x) \longrightarrow b$. Apply the same valuation calculation to the inverse. The resulting two affine maps of tail radii are inverse, so every sufficiently small target shell consists exactly of images of its corresponding source shell. Taking their unions proves the asserted punctured-disc equality. Thus the lemma is repairable; an additional assertion that $q = 1$ is unnecessary.

Definition 2.52. Let $p : A \longrightarrow \Gamma$ be a definable function. We say that p is an *$\mathfrak{o}$-partition* of A if, for every $a \in A$, the function p is constant on $\mathfrak{o}(a, p(a)) \cap A$.

This makes sense even when A has no VF-coordinates, in which case any definable function $A \longrightarrow \Gamma$ is an $\mathfrak{o}$-partition of A.

Lemma 2.53. *Let p be an $\mathfrak{o}$-partition of A. Suppose that A_{RV} is doubly bounded and A_t is closed and bounded for every $t \in A_{\mathrm{RV}}$. Then $p(A)$ is doubly bounded.*